

DLR-TomoSAR — What does the network need?

context ladder · the correlation law · amplitude vs phase

July 2026

The context ladder

TomoSAR theory inverts the *expected* covariance per pixel; no estimator is per-pixel in the data. Every practical route — the GT's own Capon included — buys that expectation with a spatial window over the stack vectors $\mathbf{g} \in \mathbb{C}^N$:

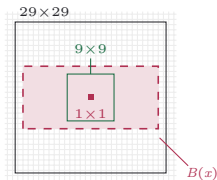
theory: one pixel, in expectation

practice: $w_a \times w_r$ boxcar $B(x)$ (az $\times$ rg)

$$\mathbf{R}(x) = \mathbb{E}[\mathbf{g}(x) \mathbf{g}^H(x)] \quad \longrightarrow \quad \hat{\mathbf{R}}(x) = \frac{1}{w_a w_r} \sum_{x' \in B(x)} \mathbf{g}(x') \mathbf{g}^H(x')$$



to scale: 9×9 sits *inside* the label's window; 29×29 is the first field to swallow it. One trunk, no downsampling, **capacity held at ~ 31 M** on every rung — the receptive field is the only thing that moves.



Why the ladder must help — the label is a window statistic

1. The label is $B(x)$ -measurable. Capon and the parameter fit are deterministic maps, so

$$y(x) = \Phi(\mathbf{g}_{B(x)}),$$

exactly, by construction — no assumption on the scene. The label holds no randomness the looks in $B(x)$ do not already hold.

3. The floor only falls. Take any two rungs with $\mathcal{F}_1 \subseteq \mathcal{F}_2$ and condition once more on the wider one:

$$\begin{aligned}\mathrm{Var}(y \mid \mathbf{g}_{\mathcal{F}_1}) &= \mathrm{E}[\mathrm{Var}(y \mid \mathbf{g}_{\mathcal{F}_2}) \mid \mathbf{g}_{\mathcal{F}_1}] \\ &\quad + \mathrm{Var}(\mathrm{E}[y \mid \mathbf{g}_{\mathcal{F}_2}] \mid \mathbf{g}_{\mathcal{F}_1}).\end{aligned}$$

The last term is a variance, hence ≥ 0 :

$$\mathrm{E}[\mathrm{Var}(y \mid \mathbf{g}_{\mathcal{F}_1})] \geq \mathrm{E}[\mathrm{Var}(y \mid \mathbf{g}_{\mathcal{F}_2})].$$

2. The error splits into a floor and a fit. For any predictor $\hat{y}$ built from a field $\mathcal{F}$,

$$\begin{aligned}\mathrm{E}[(y - \hat{y})^2] &= \mathrm{E}[\mathrm{Var}(y \mid \mathbf{g}_{\mathcal{F}})] \\ &\quad + \mathrm{E}[(\mathrm{E}[y \mid \mathbf{g}_{\mathcal{F}}] - \hat{y})^2].\end{aligned}$$

The first term is the **floor**, fixed by $\mathcal{F}$ alone; the second is what training buys.

4. And it vanishes. Once the field holds the whole window, the label's own looks are among those handed to the model:

$$\begin{aligned}B(x) \subseteq \mathcal{F} &\implies \mathbf{g}_{B(x)} \subseteq \mathbf{g}_{\mathcal{F}} \\ &\implies y = \Phi(\mathbf{g}_{B(x)}) \text{ fixed by } \mathbf{g}_{\mathcal{F}} \\ &\implies \mathrm{Var}(y \mid \mathbf{g}_{\mathcal{F}}) = 0.\end{aligned}$$

The floor of any field — the fluctuation of the looks it misses

Take *any* field $\mathcal{F}$ and split the label's window into the looks the field delivers and the looks it misses:

$$w_a w_r \hat{\mathbf{R}}(x) = \begin{cases} \sum_{p \in B(x) \cap \mathcal{F}} \mathbf{g}(p) \mathbf{g}^H(p) \\ + \sum_{p \in B(x) \setminus \mathcal{F}} \mathbf{g}(p) \mathbf{g}^H(p). \end{cases}$$

The first sum is delivered, the second is missed. Which pixels fall in which is the *only* thing that changes along the ladder. Condition on $\mathbf{g}_{\mathcal{F}}$. Every look in the first sum is one the field hands over, so that term is a constant and carries no variance — only the misses fluctuate:

$$\begin{aligned} \text{Var}(\hat{\mathbf{R}}(x) \mid \mathbf{g}_{\mathcal{F}}) &= \text{Var}\left(\frac{1}{w_a w_r} \sum_{p \in B(x) \cap \mathcal{F}} \mathbf{g}(p) \mathbf{g}^H(p) \right. \\ &\quad \left. + \frac{1}{w_a w_r} \sum_{p \in B(x) \setminus \mathcal{F}} \mathbf{g}(p) \mathbf{g}^H(p) \mid \mathbf{g}_{\mathcal{F}}\right) \\ &= \boxed{\frac{1}{(w_a w_r)^2} \text{Var}\left(\sum_{p \in B(x) \setminus \mathcal{F}} \mathbf{g}(p) \mathbf{g}^H(p) \mid \mathbf{g}_{\mathcal{F}}\right)} \end{aligned}$$

Exact: no assumption on the scene, no independence between looks. And when $B(x) \setminus \mathcal{F} = \emptyset$ the sum is empty — the floor is 0.

The three rungs — only one term moves

	pixel MLP	CNN small	CNN big
	$r = 1$	$r < w_r$	$r \geq w_a$
misses $ B(x) \setminus \mathcal{F} $	$w_a w_r - 1$	$w_a w_r - r^2$	0
label variance $E[\text{Var}(y \mid \mathbf{g}_{\mathcal{F}})]$	the misses fluctuate	the misses fluctuate	0
model error $E[(E[y \mid \mathbf{g}_{\mathcal{F}}] - \hat{y})^2]$	training buys	training buys	training buys
total $E[(y - \hat{y})^2]$	floor + model	floor + model	model alone

Every column is the same quantity, $w_a w_r - |B(x) \cap \mathcal{F}|$ — the control meets the window in a single pixel, the small CNN in all r^2 of its field, the big CNN in the whole of it. Only the shaded row moves, and no architecture or loss touches it; model error reads the same in every column, and capacity is held fixed across the ladder, so the field is the only variable.

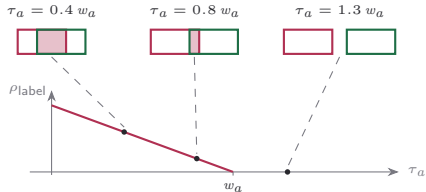
Ladder result: context grounds the prediction

metric field $\mathcal{F}$ / looks seen	pixel MLP 1×1 / 1	CNN small (Δ) 9×9 / 81	CNN big (Δ) 29×29 / 312		
curve $R^2 \uparrow$	0.24	0.47	+0.23	0.59	+0.12
per-pixel R^2 (median) $\uparrow$	-0.03	0.42	+0.45	0.57	+0.15
SSIM elevation $\uparrow$	0.918	0.949	+0.031	0.958	+0.009
SSIM range $\uparrow$	0.869	0.917	+0.048	0.928	+0.011
SSIM azimuth $\uparrow$	0.895	0.934	+0.039	0.944	+0.010
peak error median [units] $\downarrow$	2.01	0.67	-1.34	0.67	0.00
matched recall $\uparrow$	0.61	0.69	+0.08	0.68	-0.01
2 scatterers $\uparrow$	0.55	0.63	+0.08	0.70	+0.06
count exact $\uparrow$	0.56	0.68	+0.12	0.62	-0.06
matched μ MAE [units] $\downarrow$	3.37	2.37	-1.00	2.12	-0.25
2 scatterers $\downarrow$	5.70	4.30	-1.41	3.79	-0.51
matched σ MAE $\downarrow$	1.21	0.86	-0.35	0.84	-0.02
2 scatterers $\downarrow$	2.01	1.59	-0.42	1.49	-0.10
matched a MAE $\downarrow$	0.531	0.326	-0.205	0.334	+0.008

runs: conv head · sorted-GT · $K=2$ · param-MSE · active-norm · no aug (model = column). Looks seen out of $|B(x)| = w_a w_r = 26 \times 12 = 312$. Held-out test, 5.0 M px; **bold** = best, marked only where best and worst differ by more than 5%; Δ vs previous tier (**better** / **worse**); **red value** = the 1×1 control regressing its conditional mean

The correlation law — how labels share their inputs

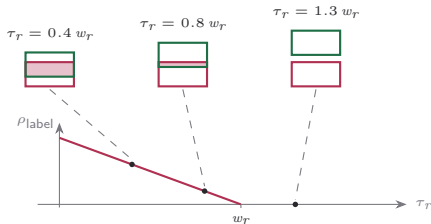
Azimuth: slide the second window along track.



The shaded overlap *is* the correlation — each case drops onto the triangle. Same law on both axes, each dying at its own w .

- **The label:** a $w_a \times w_r$ boxcar covariance around its pixel (az×rg) — every pixel of $B(x)$ enters the label at x , and nothing outside $B(x)$ does.

Range: the same construction, across track.



- **The correlation law:** two labels share input pixels in proportion to how far their boxcars overlap. The boxcar is a product of two intervals, so the law is separable:

$$\rho_{\text{label}}(\tau_a, \tau_r) = (1 - \tau_a/w_a)_+ (1 - \tau_r/w_r)_+,$$

vanishing once $\tau_a = w_a$ *or* $\tau_r = w_r$: past either the windows are disjoint and the labels share nothing.

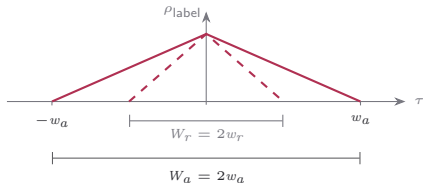
Reading the reach — the network estimates statistics

two labels at offset Δ split into shared and private looks:

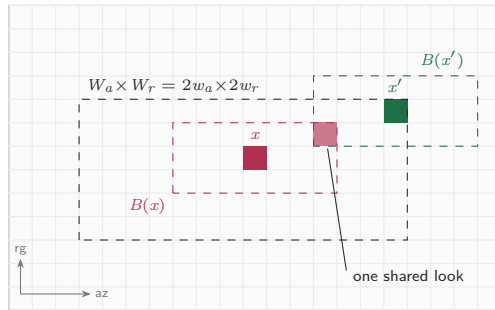
$$w_a w_r \hat{\mathbf{R}}(x) = \underbrace{\sum_{\mathcal{W} \cap \mathcal{W}'} y_p y_p^H}_{\text{shared } \mathbf{S}(\Delta)} + \underbrace{\sum_{\mathcal{W}' \setminus \mathcal{W}} y_p y_p^H}_{\text{private to } x}$$

$$w_a w_r \hat{\mathbf{R}}(x') = \underbrace{\sum_{\mathcal{W} \cap \mathcal{W}'} y_p y_p^H}_{\text{shared } \mathbf{S}(\Delta)} + \underbrace{\sum_{\mathcal{W}' \setminus \mathcal{W}} y_p y_p^H}_{\text{private to } x'}$$

the shared fraction at every offset τ is the correlation law:

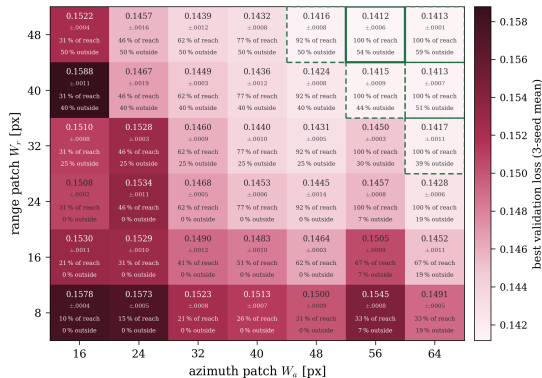


the label correlates over $\pm w_a$ in azimuth and $\pm w_r$ in range
— the centred patch spanning both triangles is $W_a \times W_r$.



x' sits at the far corner of the patch: $B(x')$ keeps a single shared look with $B(x)$ — one step further and the windows are disjoint ($|\Delta_a| = w_a$ or $|\Delta_r| = w_r$).

The verdict — saturation at the reach, axis by axis



	registered	observed
azimuth knee	$2w_a = 52$	flat past 48 — the step-8 grid brackets 52
range knee	$2w_r = 24$	40; ~ 0.003 drift 24→40, ≤ 0.0008 past 40

runs: patch $W_a \times W_r = \text{cell}$, all else fixed (full input stack, 26×12 boxcar labels); three seeded replicas per cell, cell = mean best val loss $\pm$ seed std. **Green** = best cell, dashed = inside its combined seed noise. % of reach = share of the centred $2w_a \times 2w_r$ box (52×24) the patch covers; % outside = share of the patch beyond that box. The plateau is the reach corner ($W_a \geq 48$, $W_r \geq 32$); the residual range drift past $2w_r$ clears seed noise at $W_a \geq 48$ — a third of the azimuth climb.

The price of reach — one grid against a pyramid

The trunk — one grid.

$$r = 1 + 2B(k-1)$$

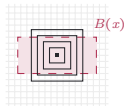
+ $2(k-1)$ per block

$$P \propto Bk^2m^2$$

capacity pinned

$$m \propto \sqrt{P/B}$$

depth paid in width



four blocks, + $2(k-1)$ each

The pyramid — halve, conv.

$$r = 1 + 2(k-1)(2^{D+1}-1)$$

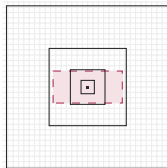
$2^\ell \times$ per station

$$m_\ell = 2^\ell m_0$$

width doubles down

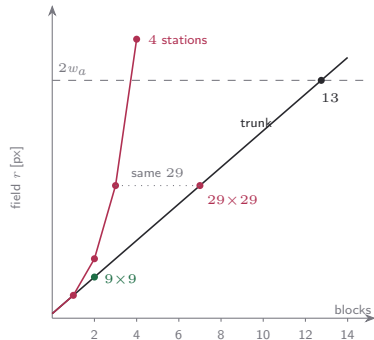
$$\frac{k^2 m_\ell^2}{4^\ell} = k^2 m_0^2$$

cost per pixel flat



four stations, the step doubling

r : field side · B : blocks stacked · m : width in channels · P : parameters · station ℓ : the block ℓ halvings down, width m_ℓ · D : deepest station

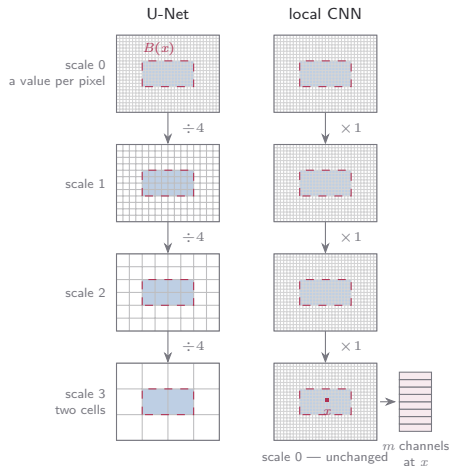
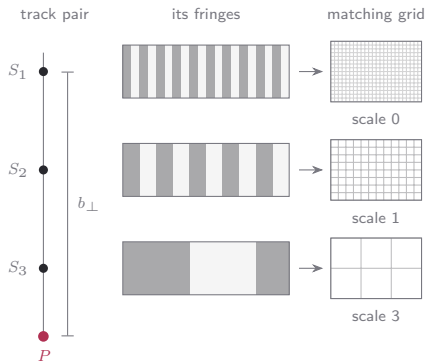


Encoder stations alone — the decoder only pushes further. Three stations hold the ladder's 29×29 ; the fourth passes the sweep's reach $2w_a$, thirteen full-resolution blocks before the trunk does.

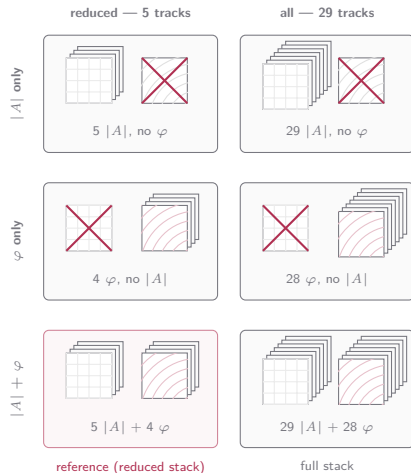
One grid per scale — matching the scales of the interferograms

Multi-scale by construction — each track pair (P, S_n) sees the same relief $z(x)$ at its own fringe rate:

$$\varphi_n(x) = (k_z^{(n)} - k_z^{(P)}) z(x) = k_n z(x), \quad k_n \propto b_{\perp, n}$$



Input ablation — which channels, and how many tracks?



Theory says the elevation information lives in the **interferogram phase**: amplitudes alone should not determine the profile. But the context ladder just showed the network leans on spatial structure — so two axes, crossed:

- **The grid**: *which channels* ($|A|$ only / φ only / both) $\times$ *how many tracks* (the reduced 5-track selection / all 29). Six retrains of the reference U-Net — same $K=2$ target, loss and split, only the input stack changes.
- **H1 — phase is everything**: strip the phase and reconstruction collapses toward nothing usable; amplitude diversity cannot buy it back.
- **H2 — priors carry part of it**: amplitude-only lands in between, and 29 $|A|$ closes part of the gap to the reference.
- **H3 — tracks pay everywhere**: if tracks are information, every modality should improve reduced $\rightarrow$ all; the open question is whether diversity can *substitute* for a missing modality.

The verdict — every channel pays on the curve, phase still places the scatterers

metric	$ A $ only		φ only		$ A + \varphi$	
	red.	all	red.	all	red.	all
<i>training (validation)</i>						
val loss ↓	0.179 ±.001	0.164 ±.002	0.162 ±.001	0.137 ±.002	0.142 ±.001	0.122 ±.000
<i>reconstruction (curve)</i>						
curve R^2 ↑	0.369 ±.014	0.437 ±.010	0.409 ±.018	0.503 ±.007	0.569 ±.008	0.667 ±.015
curve MAE ↓	0.066 ±.001	0.060 ±.001	0.059 ±.002	0.051 ±.001	0.050 ±.001	0.043 ±.001
PSNR [dB] ↑	48.7 ±.1	49.2 ±.1	48.9 ±.1	49.7 ±.1	50.3 ±.1	51.4 ±.2
SSIM elev ↑	0.946 ±.001	0.951 ±.001	0.952 ±.001	0.959 ±.001	0.959 ±.001	0.963 ±.001
SSIM range ↑	0.901 ±.001	0.911 ±.002	0.916 ±.002	0.927 ±.002	0.926 ±.002	0.933 ±.001
SSIM azim ↑	0.925 ±.000	0.933 ±.001	0.936 ±.002	0.946 ±.001	0.946 ±.001	0.953 ±.001
profile cos med ↑	0.665 ±.020	0.737 ±.017	0.912 ±.012	0.931 ±.009	0.850 ±.024	0.890 ±.012
<i>detection (matched)</i>						
precision ↑	0.758 ±.008	0.785 ±.009	0.830 ±.007	0.853 ±.008	0.821 ±.008	0.839 ±.005
recall ↑	0.652 ±.027	0.679 ±.028	0.817 ±.021	0.765 ±.038	0.641 ±.029	0.636 ±.024
matched F1 ↑	0.701 ±.019	0.728 ±.020	0.823 ±.012	0.806 ±.024	0.720 ±.020	0.723 ±.017
count exact ↑	0.609 ±.034	0.633 ±.035	0.802 ±.029	0.741 ±.048	0.595 ±.034	0.591 ±.033
peak err med ↓	1.48 ±.30	1.34 ±.00	0.67 ±.00	0.67 ±.00	0.81 ±.30	0.81 ±.30
peak err p95 ↓	15.6 ±.3	15.3 ±.3	11.7 ±.1.9	14.0 ±.3	15.4 ±.5	15.2 ±.4

runs: UNet · conv · sorted-GT · $K=2$ · param-MSE · presence none · full norm ladder · hv flips · patch 64×32 (input = column); five seeded replicas per arm, cells mean ± seed std; held-out test, 5.0 M px. red. = 5 tracks (5 $|A|$ / 4 φ); all = 29 tracks (29 $|A|$ / 28 φ). **bold** = best in row, marked only where best and worst differ by more than 5%; **green** = the full-stack curve peak and the phase detection wins, **red** = amplitude diluting recall; matched = Hungarian pairing. Peak err med/p95: per-pixel peak-error distribution; profile cos med: median per-pixel cosine to the GT profile.

Input ablation by scatterer count — each parameter reads its own modality

metric	$ A $ only		φ only		$ A + \varphi$	
	red.	all	red.	all	red.	all
matched precision $\uparrow$	0.758 $\pm$.008	0.785 $\pm$.009	0.830 $\pm$.007	0.853 $\pm$.008	0.821 $\pm$.008	0.839 $\pm$.005
$k=1$	0.803 $\pm$.012	0.821 $\pm$.011	0.875 $\pm$.007	0.890 $\pm$.009	0.853 $\pm$.008	0.867 $\pm$.011
$k=2$	0.696 $\pm$.009	0.734 $\pm$.009	0.761 $\pm$.012	0.803 $\pm$.008	0.782 $\pm$.007	0.806 $\pm$.007
matched recall $\uparrow$	0.652 $\pm$.027	0.679 $\pm$.028	0.817 $\pm$.021	0.765 $\pm$.038	0.641 $\pm$.029	0.636 $\pm$.024
$k=1$	0.684 $\pm$.046	0.704 $\pm$.047	0.882 $\pm$.040	0.779 $\pm$.065	0.620 $\pm$.046	0.593 $\pm$.043
$k=2$	0.606 $\pm$.009	0.643 $\pm$.007	0.725 $\pm$.009	0.745 $\pm$.003	0.672 $\pm$.008	0.698 $\pm$.008
count acc pred $k=1 \uparrow$	0.961 $\pm$.004	0.963 $\pm$.004	0.969 $\pm$.005	0.964 $\pm$.003	0.948 $\pm$.007	0.959 $\pm$.002
count acc pred $k=2 \uparrow$	0.646 $\pm$.003	0.665 $\pm$.007	0.724 $\pm$.007	0.770 $\pm$.005	0.733 $\pm$.010	0.767 $\pm$.007
matched μ MAE $\downarrow$	2.95 $\pm$.09	2.65 $\pm$.08	2.29 $\pm$.08	2.04 $\pm$.12	2.30 $\pm$.08	2.15 $\pm$.06
$k=1$	1.25 $\pm$.05	1.03 $\pm$.04	0.55 $\pm$.05	0.51 $\pm$.02	0.59 $\pm$.04	0.52 $\pm$.02
$k=2$	4.88 $\pm$.08	4.52 $\pm$.06	4.58 $\pm$.15	3.86 $\pm$.12	4.05 $\pm$.08	3.74 $\pm$.06
matched σ MAE $\downarrow$	0.97 $\pm$.03	0.91 $\pm$.03	0.89 $\pm$.04	0.79 $\pm$.05	0.87 $\pm$.03	0.80 $\pm$.03
$k=1$	0.29 $\pm$.02	0.26 $\pm$.01	0.24 $\pm$.01	0.20 $\pm$.01	0.25 $\pm$.01	0.21 $\pm$.01
$k=2$	1.74 $\pm$.01	1.64 $\pm$.01	1.76 $\pm$.05	1.51 $\pm$.05	1.51 $\pm$.01	1.38 $\pm$.01
matched a MAE $\downarrow$	0.369 $\pm$.017	0.356 $\pm$.015	0.464 $\pm$.009	0.446 $\pm$.025	0.374 $\pm$.026	0.346 $\pm$.012
$k=1$	0.164 $\pm$.015	0.161 $\pm$.010	0.233 $\pm$.010	0.234 $\pm$.021	0.161 $\pm$.017	0.165 $\pm$.016
$k=2$	0.601 $\pm$.007	0.579 $\pm$.027	0.771 $\pm$.013	0.698 $\pm$.011	0.593 $\pm$.022	0.524 $\pm$.007

runs: UNet · conv · sorted-GT · $K=2$ · param-MSE · presence none · full norm ladder · hv flips · patch 64×32 (input = column); held-out test, cells mean $\pm$ seed std over five seeded replicas; k = true scatterer count of the pixel. red. = 5 tracks, all = 29 tracks. **bold** = best in row, marked only where best and worst differ by more than 5%; **green** = detection and position living in the phase, **red** = amplitude error without the $|A|$ channels. Precision rewards under-prediction — rank on recall. count acc | pred k : share of pixels claiming exactly k scatterers whose true count is k . μ , σ in elevation units; a in linear amplitude.

Input ablation, predicted statistics vs the GT — phase fires the slots, the full stack calibrates layover

		A only		φ only		A + φ	
prediction mean	GT	red.	all	red.	all	red.	all
slot 0 — strongest scatterer; GT-active on every pixel							
active frac	1.00	0.74 $\pm$.03	0.75 $\pm$.04	0.90 $\pm$.03	0.82 $\pm$.05	0.69 $\pm$.03	0.67 $\pm$.03
a — active	0.77	0.85 $\pm$.03	0.83 $\pm$.04	0.71 $\pm$.04	0.72 $\pm$.02	0.90 $\pm$.03	0.92 $\pm$.04
μ — active	−4.89	−4.03 $\pm$.10	−4.15 $\pm$.17	−4.29 $\pm$.05	−4.52 $\pm$.08	−3.92 $\pm$.10	−4.16 $\pm$.09
σ — active	1.76	1.49 $\pm$.04	1.50 $\pm$.03	1.41 $\pm$.05	1.59 $\pm$.06	1.48 $\pm$.05	1.65 $\pm$.04
slot 1 — second scatterer; GT-active on 26% of pixels							
active frac	0.260	0.346 $\pm$.003	0.338 $\pm$.006	0.338 $\pm$.006	0.308 $\pm$.004	0.299 $\pm$.006	0.289 $\pm$.004
a — active	1.43	1.05 $\pm$.02	1.08 $\pm$.05	1.05 $\pm$.05	1.08 $\pm$.03	1.04 $\pm$.03	1.15 $\pm$.01
a — inact.	0.00	0.68 $\pm$.01	0.64 $\pm$.03	0.44 $\pm$.04	0.48 $\pm$.01	0.42 $\pm$.01	0.49 $\pm$.03
μ — active	11.34	9.73 $\pm$.46	9.60 $\pm$.61	10.20 $\pm$.14	10.25 $\pm$.51	10.09 $\pm$.33	10.79 $\pm$.29
μ — inact.	—	4.04 $\pm$.38	2.72 $\pm$.33	3.51 $\pm$.35	2.55 $\pm$.28	4.09 $\pm$.31	3.71 $\pm$.21
σ — active	3.53	3.10 $\pm$.03	3.12 $\pm$.05	2.99 $\pm$.02	3.11 $\pm$.05	2.94 $\pm$.05	3.18 $\pm$.03
σ — inact.	—	2.47 $\pm$.02	2.19 $\pm$.04	1.91 $\pm$.02	1.86 $\pm$.02	1.87 $\pm$.03	1.81 $\pm$.02
distribution width — std over GT-active pixels							
a — slot 0	1.48	1.15 $\pm$.02	1.16 $\pm$.05	1.00 $\pm$.06	1.00 $\pm$.02	1.17 $\pm$.03	1.19 $\pm$.01
μ — slot 0	3.30	1.73 $\pm$.08	1.91 $\pm$.07	2.11 $\pm$.07	2.33 $\pm$.10	1.99 $\pm$.07	2.24 $\pm$.07
σ — slot 0	2.84	0.83 $\pm$.01	0.91 $\pm$.02	1.01 $\pm$.05	1.35 $\pm$.07	0.95 $\pm$.03	1.28 $\pm$.01
a — slot 1	1.45	0.77 $\pm$.02	0.83 $\pm$.05	0.86 $\pm$.04	0.87 $\pm$.02	0.83 $\pm$.04	0.89 $\pm$.01
μ — slot 1	16.24	7.14 $\pm$.29	7.74 $\pm$.36	9.37 $\pm$.31	10.90 $\pm$.75	8.72 $\pm$.18	10.06 $\pm$.27
σ — slot 1	4.01	0.80 $\pm$.03	0.92 $\pm$.04	1.08 $\pm$.02	1.33 $\pm$.04	1.07 $\pm$.02	1.34 $\pm$.03

runs: UNet · conv · sorted-GT · $K=2$ · param-MSE · presence none · full norm ladder · hv flips · patch 64×32 (input = column); five seeded replicas per arm, cells mean $\pm$ seed std; means conditioned on the slot's GT activity mask. red. = 5 tracks, all = 29 tracks. **bold** = closest to the GT column, marked only where best and worst differ by more than 5%; **green** = phase firing the slots and the full-stack slot-1 calibration. Inactive GT slots define no μ/σ reference (—). Width rows: std of the prediction over GT-active pixels vs the GT's own std. μ , σ in elevation units; a in linear amplitude.